AZURE DAY 2026 HACKATHON · DISASTER RECOVERY AGENT

Disaster Recovery (DR) Agent

Autonomous Disaster Recovery Validation

Built 100% on Azure. AI agents that validate DR by executing real tests: fail over, measure RTO/RPO, fail back, and prove you can recover, on any day instead of once a year.

Fail over **Measure** **Fail back** **Prove**

Team Azure Armada · Virtusa

Downtime now costs about $900,000 an hour

The latest independent research (2026) and a 2025 cloud outage: the cost of not recovering on time.

$15,000

per minute of downtime

is the average for Global 2000 companies: about $900,000 an hour and $600B a year in total. Regulatory fines alone average $51M per company a year.

Splunk & Oxford Economics, The Hidden Costs of Downtime, May 2026 (2,000 executives) [1]

57%

had a last major outage over $100K

and 1 in 5 operators said their last major outage cost more than $1 million, for the second year running.

Uptime Institute, Annual Outage Analysis 2026 [2]

$500–650M

from one cloud-region outage

lost by US companies when AWS us-east-1 failed on 20 October 2025. Even hyperscalers fail; what matters is whether your recovery works.

Parametrix estimate, Oct 2025 [3]

At $15,000 a minute, one hour of downtime costs about $900,000. Recovery that is assumed, not proven, is an unpriced risk.

In India, failing to recover is punished by freezing growth

Regulators act on outages and weak disaster recovery, not just on breaches.

HDFC Bank — Dec 2020

After repeated outages, the latest caused by a power failure at a primary data centre, the RBI stopped all new digital launches and new credit-card sourcing.

≈15 months

until digital launches were allowed again (Mar 2022); cards resumed Aug 2021

[4]

NSE — Feb 2021

Trading was halted for nearly four hours on 24 February 2021. NSE and NSE Clearing settled with SEBI; executives were ordered to do community service and training.

≈ $8.8M

paid by NSE and NSE Clearing to settle with SEBI (2023)

[5]

Kotak Mahindra Bank — Apr 2024

The RBI cited deficiencies including business continuity and disaster recovery, and frequent outages, and barred online customer onboarding and new credit cards.

≈10 months

until the restrictions were lifted (Feb 2025)

[6]

For a bank, a 10–15 month ban on new digital customers costs far more than any fine.

Regulators now require proof of recovery, on a schedule

Having a DR plan is not enough. It must be tested, the gaps fixed, and the test repeated.

Regulator	What it requires	If you fall short
RBI **Cybersecurity, Technology: Risk, Resilience and Assurance Framework Directions, 31 Jul 2026**	DR drills for critical information systems at least half-yearly. Any major issue found must be fixed and re-tested successfully before the next cycle. [7]	Supervisory restrictions on business, as with HDFC Bank and Kotak Mahindra Bank. [4][6]
SEBI **Cybersecurity & Cyber Resilience Framework**	Resume critical operations within 2 hours (RTO) with a 15-minute RPO. Market infrastructure: run from the DR site within 45 minutes of declaring a disaster, drills every quarter, unannounced live trading from the DR site every six months. [8][9]	Enforcement and settlements, as with NSE: about $8.8M. [5]
EU DORA **Regulation (EU) 2022/2554, applies from 17 Jan 2025**	Test ICT business continuity and response and recovery plans at least yearly and after major changes (Art. 11(6)); test backup restoration periodically (Art. 12). [10]	Penalties set by each EU state (Art. 50). Critical ICT providers: up to 1% of average daily worldwide turnover, every day, for up to six months (Art. 35). [10]

Our demo uses the same yardstick: target RTO 30 min and RPO 15 min, stricter than SEBI's 2 hours, and every drill produces timestamped, audit-ready evidence.

Yet most organisations cannot prove their recovery works today

Plans exist on paper. Tests are rare, manual and partial, so the gaps surface during the real disaster.

< 1 in 3

organisations perform regular failover testing

Cockroach Labs, State of Resilience 2025 (1,000 executives) [11]

20%

of executives believe their organisation is fully prepared to prevent or respond to outages

Cockroach Labs, State of Resilience 2025 [11]

#1

cause of human-error outages: staff not following established procedures

Uptime Institute, Annual Outage Analysis 2026 [2]

Why the gap persists

Tests are disruptive and manual

A real failover needs several teams, a change window and hand-written timings, so it happens rarely.

Environments drift between tests

A setting added in one region but not the other only shows up when you fail over.

Runbooks go stale

Documents name servers and services that no longer exist; engineers find out mid-incident.

RTO and RPO are assumed

Targets are signed off on paper; actual recovery time and data loss are rarely measured end to end.

What an unproven DR plan costs: one bad morning

Illustration for a tier-1 payments service, using the cited figures. Not a forecast; the inputs are shown.

Recovery target (RTO)	**30 minutes**	Signed off on paper; never measured end to end
Actual recovery	**4 hours**	Comparable to the NSE trading halt of Feb 2021 [5]
Hourly cost of downtime	**$900,000**	$15,000 a minute, the Global 2000 average (Splunk & Oxford Economics, 2026) [1]
Direct loss	**$3.6M**	4 h × $900K; $3.15M of it beyond the 30-minute target
Regulatory precedent	**≈ $8.8M**	What NSE paid to settle its 2021 outage case [5]; banks have faced 10–15 month business restrictions [4][6]

The DR Agent measured 34.35 s to recover the same kind of service in our live Azure drill (18 Oct 2026), with zero payments lost, and proves it again on every run.

The DR Agent runs the real drill and proves the result

Six AI agents on Azure AI Foundry plan and judge; a guarded engine executes on live Azure resources; humans approve.

Phase 1

Design & prepare

Architect — picks the cheapest strategy that truly meets RTO/RPO, with real Azure prices

Readiness — 7 live checks; finds drift before the disaster; one-click approved fix

Runbook — flags stale steps and writes the corrected runbook

Phase 2

Plan, execute, fail back, report

Planner — builds a 21-step drill from an allowlist of safe actions

Engine + Validator — injects the outage, fails over, measures; diagnoses any failure live

Reporter — audit-ready report with every second of RTO accounted for

	Today	With the DR Agent
How often DR is proven	Once or twice a year	On demand, any day
RTO and RPO	Assumed on paper	Measured on every drill, to the millisecond
Configuration drift	Found during the real outage	Found by readiness checks, fixed with approval
Audit evidence	Screenshots and spreadsheets	Generated report with timestamps and approvals

Six specialist agents, each with one job and a contract

Versioned prompt agents in Azure AI Foundry Agent Service. Every answer must match a strict JSON schema and pass code checks.

Agent	Model	Gets (facts from code)	Returns	Checked by code
Architect	`gpt-4.1-mini`	Profile, inventory, live Azure prices	Strategy + all 5 options	Cheapest option that truly meets targets
Readiness	`gpt-4.1-mini`	7 deterministic checks with evidence	GO / NO-GO, risks	Any failed check forces NO-GO
Runbook	`gpt-4.1-mini`	Runbook markdown, inventory	Stale / valid per step, corrected runbook	Schema only (advisory)
Planner	`gpt-5-mini`	Strategy, risks, allowlist, safe sequence	21-step drill plan	Allowlisted actions, mandatory sequence, approvals
Validator	`gpt-5-mini`	Failed step, error, events, live probes	Cause, evidence, fix	Fix allowlisted; never Region A mid-outage; human approves
Reporter	`gpt-4.1-mini`	Measured RTO/RPO, timeline, findings	Summary, remediation items	Numbers rendered by code, not retyped

Watch them think. A live activity panel streams every Azure read, the Foundry call (agent version, latency, tokens) and the code check, as it happens. It caught a real defect: the Architect compared a cost multiplier with a dollar budget; we now give it Azure list prices and it is right first time (5 of 5 runs).

Agents decide. The engine acts. Humans approve.

DR touches production, so no model can call an Azure write API. Every action is allowlisted, ordered, approved and reversible.

Allowlisted actions

The engine runs only 15 known actions (stop/start app, SQL failover, endpoint on/off, ASR test). Anything else is rejected.

Mandatory safe sequence

Every plan must contain heartbeat → outage → SQL failover → reroute → verify → measure → fail back, in that order.

Human approval gates

The plan, the forced database failover, the failback and any Validator fix wait for a named approver.

Kill switch and rollback

One click stops the drill; rollback restores the primary region and the database role.

Fallback for every agent

Foundry agent → model deployment → deterministic rules. The screen shows which one answered.

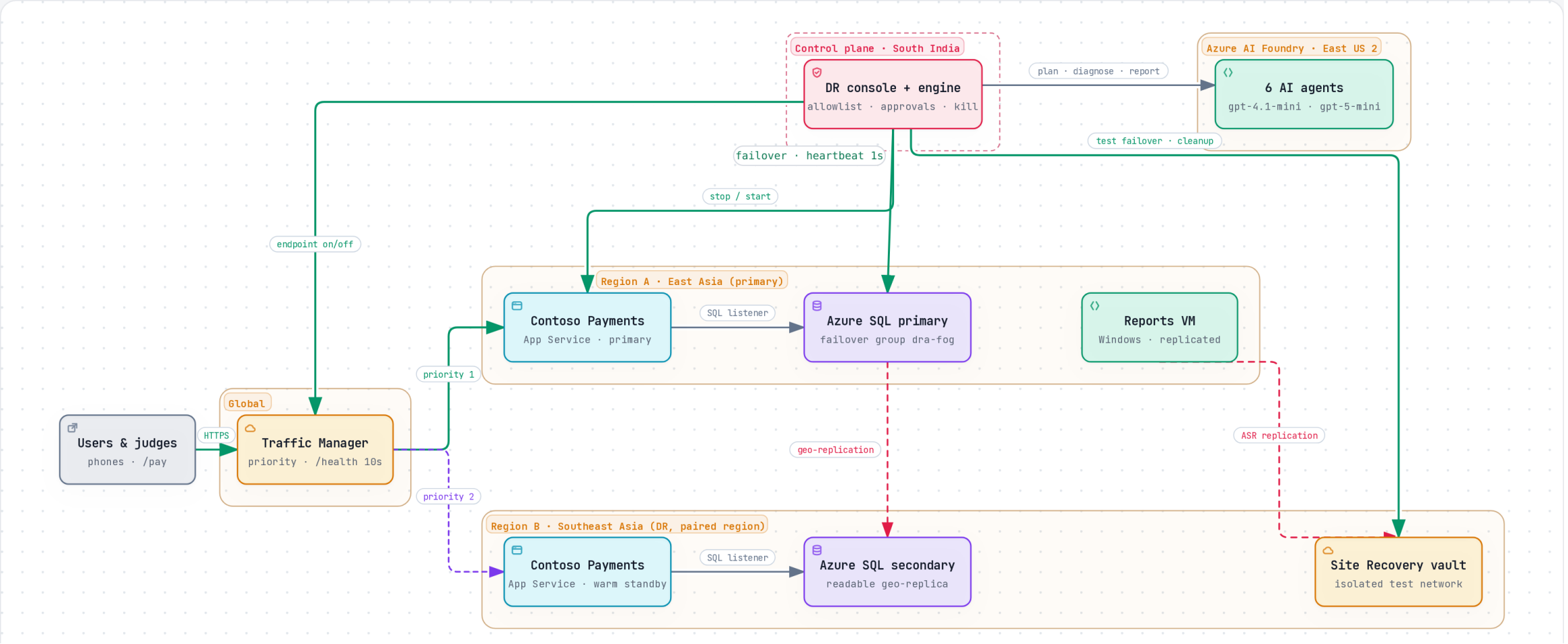
Full audit trail

Every event, approval and measurement is timestamped and stored, then written into the report.

Proven in a chaos drill: we disabled the DR traffic endpoint mid-failover. The Validator diagnosed it from live probes, proposed the allowlisted fix, a human approved, and recovery completed in 57.4 s.

Architecture: 100% Azure, across an Azure paired region

East Asia (primary) and Southeast Asia (DR). The control plane runs outside both, in South India, so it survives the disaster it manages.



App Service ×2 | Azure SQL failover group | Traffic Manager | Site Recovery | AI Foundry Agent Service | Key Vault + managed identity | Storage | Monitor + App Insights | Bicep: one-click up / down

Measured on live Azure, not assumed

Real failovers of the payments service, East Asia → Southeast Asia, 17–18 Oct 2026. Target: RTO 30 min, RPO 15 min.

34.35 s

RTO, latest drill

29.6–40.4 s across 8 drills (median 33.5 s)

0 s

RPO, every drill

no confirmed payment lost (1 s heartbeat proof)

3 min 14 s

Failback to primary

planned, no data loss

5 min 45 s

VM recovered by Site Recovery

proven by a command inside the VM

Where the 34.35 s went (latest drill, one clock, segments add up exactly)



Outage injected 0.519 s

SQL forced failover 15.355 s

Traffic Manager reroute 5.916 s

DNS + health checks reach DR 11.619 s

Engine bookkeeping 0.944 s

Why judges can trust the numbers: RTO runs from the injected outage to the first healthy check through Traffic Manager (1 s probes; the previous failed check is shown as the precision window). RPO compares every confirmed heartbeat write with the rows that survived in the new primary. Chaos drill with a surprise fault: 57.4 s.

What it costs to prove recovery

Actual Azure bills and list prices for what we built, compared with the cost of downtime.

$0.09

all AI agent calls

every Foundry agent run across 2 days of building and 9 drills: $0.087 (Azure Cost Management) [12]

$2.44

entire environment so far

two regions, database, VM, console, monitoring: everything, 17–18 Oct 2026 (Azure Cost Management) [12]

$38.26

per month for the DR set-up

both regions + Traffic Manager at Azure list prices (Retail Prices API) [13]

A month of this DR set-up costs less than a fifth of a second of downtime at the 2026 average of $15,000 a minute ($38.26 ÷ $250 per second = 0.15 s). [1][13]

Cost-aware by design: the Architect prices every strategy from live Azure list prices and flags anything over budget; the whole environment is created and removed with one command (up.sh / down.sh), so nothing runs when it is not needed.

Live demo, and where it goes next

What you will see in the next minutes, on your own phone.

The demo

1. Scan the QR code: Contoso Payments opens on your phone; make a payment
2. Readiness says NO-GO: Southeast Asia is missing a setting; one approved click fixes it
3. The Planner builds the drill; we approve it and inject the outage
4. Your phone goes red, then reconnects to Southeast Asia in ~30 s with the same balance
5. Failback, then the report: RTO, RPO and every second accounted for

Roadmap (not in the demo)

Scheduled drills

weekly, and after every infrastructure change

More services

AKS, Cosmos DB, Storage, Service Bus as new allowlisted actions

Azure Chaos Studio

standard fault library alongside our own faults

Approvals where teams work

Microsoft Teams approval cards

Portfolio view

resilience score across many applications

Sources

Every figure in this deck is taken from the source below. Our own figures come from live drills on Azure.

[1] Splunk with Oxford Economics, The Hidden Costs of Downtime 2026, 19 May 2026 (2,000 Global 2000 executives). newsroom.cisco.com (The $600 Billion Wake-up Call)

[2] Uptime Institute, Annual Outage Analysis 2026, press release 13 May 2026. uptimeinstitute.com

[3] Parametrix estimate of losses from the AWS us-east-1 outage of 20 Oct 2025 (Insurance Edge, 30 Oct 2025).

[4] RBI order to HDFC Bank, 2 Dec 2020 (Outlook Business, 3 Dec 2020); curbs lifted 17 Aug 2021 (cards) and 11 Mar 2022 (digital) (Business Standard, 12 Mar 2022).

[5] NSE and NSE Clearing settle the 24 Feb 2021 trading-halt case with SEBI for ₹72.64 crore (Moneylife; Business Standard, Jun 2023); ≈ $8.8M at June 2023 rates of ₹82–83 per US$.

[6] RBI action on Kotak Mahindra Bank under Sec. 35A, 24 Apr 2024 (TechCrunch); restrictions lifted 12 Feb 2025 (Business Standard).

[7] RBI, Cybersecurity, Technology: Risk, Resilience and Assurance Framework Directions, 31 Jul 2026, summarised by KPMG India, Sep 2026.

[8] SEBI Cybersecurity and Cyber Resilience Framework: RTO 2 hours, RPO 15 minutes (clarification reported by PTI / The Week, 28 Aug 2025).

[9] SEBI framework for market infrastructure institutions: DR site within 45 minutes, quarterly drills (Business Standard, 22 Mar 2021).

[10] Regulation (EU) 2022/2554 (DORA), Articles 11(6), 12, 35(7)–(8), 50, 51. eur-lex.europa.eu/eli/reg/2022/2554/oj

[11] Cockroach Labs with Wakefield Research, The State of Resilience 2025 (1,000 executives, Aug–Sep 2024). cockroachlabs.com

[12] Azure Cost Management, actual cost by service in US$ (CostUSD), month to date, queried 18 Oct 2026 (our environment).

[13] Azure Retail Prices API (prices.azure.com), East Asia / Southeast Asia list prices for App Service B1 Linux, SQL Database Basic, Traffic Manager, 18 Oct 2026.

Our measurements: 9 drills recorded by the DR Agent (17–18 Oct 2026); each report keeps the raw timestamps, events and approvals.

Don't assume you can recover.

Let the DR Agent test it for real, and prove it, whenever you need.

Real DR drills
Fail over, measure, fail back: 9 live drills, RTO ~34 s, RPO 0

Built 100% on Azure
AI Foundry, SQL failover groups, Traffic Manager, Site Recovery, Monitor

Safe by design
Allowlisted actions, human approval, kill switch, full audit trail

Thank you · Team Azure Armada